# Aswini Sathyan C

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## PROFILE

AI/ML-focused MCA student with strong foundations in Python programming, machine learning, and data analysis. Hands-on experience in computer vision using YOLO-based object detection, AI-driven automation systems, and large-scale data preprocessing pipelines. Skilled in model integration, statistical analysis, and database-backed application development, with a focus on building scalable, efficient, and real-time intelligent systems. Strong analytical mindset with collaborative teamwork and continuous learning approach.

## EDUCATION

**MCA** — 2024 – Present
*Government Engineering College , Thrissur* — Thrissur , India
*CGPA : 8.2*

**BSc Physics** — 2021 – 2024
*Sreekrishna College , Guruvayur* — Guruvayur , India
CGPA : 7.21

## TECHNICAL SKILLS

- Programming Languages: Python, Java
- Version Control & Collaboration: Git, GitHub
- Frontend Technologies: HTML, CSS
- Database : MySQL
- AI & Machine Learning: YOLOv8, model training & evaluation, image processing
- Libraries & Frameworks: Scikit-learn, TensorFlow, NumPy, Pandas, OpenCV, Matplotlib

## PROJECTS

**Green VEICART: AI & IoT-Based Smart Checkout System for Fruits and Vegetables**
- Developed an AI-based billing system using YOLOv8 object detection for real-time product identification.
- Integrated ESP32 and HX711 sensors for real-time weight measurement and automated price calculation.
- Implemented backend processing and database management using Python and SQLite.
- Built a Streamlit user interface for displaying model predictions and handling QR-based payments.
- Improved billing accuracy and efficiency using computer vision and automation.

**Identification of Lightning Hotspot Using Satellite Data and Analysis of Atmospheric Parameters**
- Processed large-scale satellite lightning datasets using Python (NumPy, Pandas, Matplotlib)
- Performed data cleaning, preprocessing, and statistical analysis on climate data
- Identified global lightning hotspot regions with focus on the Congo basin
- Analyzed correlation between lightning activity and atmospheric parameters (CAPE, K-Index)
- Generated visualizations and data-driven insights using NASA and ERA5 datasets

## LANGUAGES

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|---|---|---|---|
| English | ●●●●○ | Malayalam | ●●●●● |
| Tamil | ●●○○○ | | |

## CORE COMPETENCIES

Problem Solving | Time Management | Communication | Leadership | Analytical Thinking

## INTERESTS

Travelling | Badminton | Gardening